

Tesla Zhang

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Education

B.S. in Computer Science at **The Pennsylvania State University**, PA, US
Minor in Mathematics, GPA 3.28/4.00, selected courses: Math 435, Cmpsc 450, Math 427, Math 429, Math 437

Aug, 2018 – Dec, 2022

Ph.D. in Computer Science at **Carnegie Mellon University**, PA, US
Advisor: Stephanie Balzer, selected courses: 15-836, 15-791

Work Experience

JetBrains Research, Remote

Jan, 2020 – Dec, 2020

HoTT and Dependent Types, Interactive Theorem Prover Development

- Improved the language/IDE, such as sections, hygiene macros, Fin type with elaborative subtyping, semantic highlighting, etc.
- Created a debugger for inspecting bidirectional type-checking and REPL in both CLI and IntelliJ IDEA.

PLCT Lab, Remote

Dec, 2020 – Present

Implementation of Dependent Types, Opensource Maintainer

RisingWave Labs, Remote

Jul, 2022 – Jul, 2023

Streaming Database, Developer Intern

- Implemented a pretty printing framework for trees with smart line fitting and Unicode art. Integrated into SQL explain.

Sourcebrella Inc., Shenzhen, China

Feb, 2018 – Jul, 2018

Static Analysis, Compiler Frontend, IDE Plugin Development

PingCAP Inc., Remote

Aug, 2018 – Aug, 2019

Distributed Storage Systems, TiKV Intern – Ecosystem Team

Related Projects

Aya Prover, Practical Implementation of Dependent Types (role: project leader)

•Supports dependent

types, dependent pattern matching with confluence check for overlapping clauses, higher inductive types, GADTs (paper published), hierarchical universes, cubical type theory features, and implicit arguments.

- Can export elaboration result to HTML or LATEX. Can JIT-compile closures into JVM using HOAS, Can refine patterns using coverage information. Supports both LSP in VSCode and IntelliJ PSI. Provide jlink binary releases.

IntelliJ Pest, Pest language plugin for IntelliJ Platform

•Semantic-based

highlighting, completion, navigation, definition extraction/inlining, and Rust plugin integration.

- Provides live preview – test grammar files by dynamically highlighting user code according to the grammar on the fly. These highlighted code could be exported to HTML.

Skills

- Programming Languages: multilingual (not limited to any specific language), especially experienced in Java Kotlin Rust C# Agda Haskell Arend, comfortable with Dart C C++ F# F★ Idris Perl MATLAB (in random order).
- Compiler: understand techniques like locally nameless, explicit substitution, ANF, (P)HOAS (in LF & logic programming), and NbE.
- Kotlin/Java: **10 years of experience**, familiar with JNI, JPMS, Gradle, Kotlin coroutines, and Swing.
- Type Theory: understand Martin-Löf type theory, coinduction, HoTT, and Cubical, familiar with Idris, Agda (**5 years** of experience, contributor), Arend and some Lean/F★/Coq.
- IDE Tooling: **6 years of experience**, familiar with the IntelliJ Platform infrastructure (created , , , etc.), also have experience with VSCode plugin development.
- Tools: editor-agnostic, have experience with team tools like YouTrack, Jira, GitHub, BitBucket, Slack, JetBrains Space and more.

Misc

- Profile links (please use a PDF reader with hyperlink support): , Marketplace
- Languages: English - fluent (TOEFL 100), Chinese - native speaker
- Open-source contributions: g, contributed to **agda**, **Arend**, **libgdx**, **jacoco**, **KaTeX**, **shields.io**, **grpc-rs**, **intellij-solidity**, **intellij-haskell**, **intellij-rust**, **TeXiFy-IDEA**, **rust-analyzer** and other projects
- w: 6000+ reputation, also active on (5000+ reputation) and
- Latest revision of this resume: one-page version n, complete version: t
- 1 dan** on , ranked #111 on the whole site (Top 0.020%), primarily in Haskell, Agda, and Idris

Publications & Preprints

[1]T. Zhang, “A Simpler Encoding of Indexed Types,” in *Proceedings of the 6th ACM SIGPLAN International Workshop on Type-Driven Development*, in TyDe '21. Republic of Korea: ACM, 2021. doi: .